

Proposals for realizing a Josephson diode in atomtronic circuits

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(Received 27 November 2025; revised 2 January 2026; accepted 4 June 2026; published 23 June 2026)

The Josephson diode, a nonreciprocal quantum element analogous to the familiar semiconductor p - n junction diode, has been realized in solid-state systems. However, it remains relatively unexplored in atomtronic circuits, which, in contrast to the solid state, allow real-time, *in situ*, and tunable control over key elements such as junction geometry, current-phase relationship, symmetry breaking, and interaction-induced nonlinearity. In this Letter, we propose and numerically demonstrate the realization of the Josephson diode effect in an atomtronic circuit consisting of a ring-shaped Bose-Einstein condensate and with optical barriers serving as Josephson junctions. Our implementation of this macroscopic nonreciprocal quantum phenomenon is based on realizing the required inversion symmetry breaking through asymmetric barrier placement and an asymmetric alternating current drive, enabling position- and drive-tunable diode effects with efficiencies up to 20% and 91%, respectively. While standard time-of-flight absorption imaging can readily observe these effects, we employ cavity optomechanics for cleaner, *in situ*, real-time, and nondestructive measurements of the relevant condensate dynamics. Our results establish a highly tunable platform for nonreciprocal Josephson transport, opening avenues for diode-based neutral-atom technologies in future quantum circuits.

DOI: [10.1103/7kb2-118m](https://doi.org/10.1103/7kb2-118m)

Introduction. The semiconductor p - n junction diode, a foundational element of modern electronics, serves as a classic example of a nonreciprocal circuit element and finds application in signal processing, logic circuits, and sensors [1–3]. Inspired by the success of the p - n junction diode, a key goal has been to realize a superconducting analog that exhibits similar nonreciprocal behavior, but without the inherent dissipation due to resistance. This has led to theoretical proposals for realizing the superconducting diode effect (SDE) in Josephson junctions, termed as the Josephson diode effect (JDE) [4–7].

Experimentally, the SDE was initially observed in junction-free superconductors [8–16], followed by its realization in Josephson junctions (JJs) [17–20], and in superconducting quantum interference devices (SQUIDs) [21–25]. This effect requires the breaking of inversion symmetry (IS) and time-reversal symmetry (TRS) of the system, achieved through an external magnetic field or intrinsic spin-orbit coupling [6]. Remarkably, recent experiments have demonstrated field-free superconducting diodes that rely solely on IS breaking, achieving robust half-wave rectification over multiple cycles [26,27]. Theoretically, these effects have been widely explored using the resistively and capacitively shunted junction (RCSJ) model [28,29], Bogoliubov–de Gennes analysis [15,30–32], and Green’s function techniques [33]. A prominent physical platform for investigating these effects is the asymmetric SQUID, where a nonsinusoidal current-phase relationship (CPR) and external flux bias give rise to the nonreciprocal transport [34–37].

Beyond solid-state superconducting devices, *atomtronic circuits*, composed of Bose-Einstein condensates (BECs), offer a powerful and complementary route for exploring the Josephson diode effect. They provide a powerful analog platform with real-time, *in situ*, and tunable control over key ingredients such as junction geometry, current-phase relationship, symmetry breaking, and interaction-induced nonlinearity, enabling direct manipulation of coherent matter waves [38–41]. In typical atomtronic implementations, JJs are realized using tightly focused, repulsive optical barriers that separate a BEC into two or more weakly coupled regions [42–44]. A bias current is generated by imposing a relative motion between the barrier and the condensate [44–46]. The critical current that the barrier can support is determined by the current-chemical potential difference (I-V) relation, which is measured by counting the number of atoms in the two wells using absorption imaging [44–48]. However, these traditional imaging techniques are destructive and require repeated preparation of the BEC. Recently, we have addressed this limitation by employing cavity optomechanics to infer the chemical potential difference in an atomtronic SQUID in a nondestructive, *in situ*, real-time, single-shot manner [49], offering a major advance over the absorption imaging-based methods.

In this Letter, we theoretically propose an atomtronic Josephson diode by breaking inversion symmetry in a ring BEC using two experimentally accessible configurations: (i) asymmetrically placed JJs and (ii) a biharmonic ac drive. In the first configuration, asymmetrically placed junctions divide the ring into arms of unequal length. Under a bias current induced by the junction motion, the dynamics become direction dependent, and the dc–ac transition occurs earlier when the

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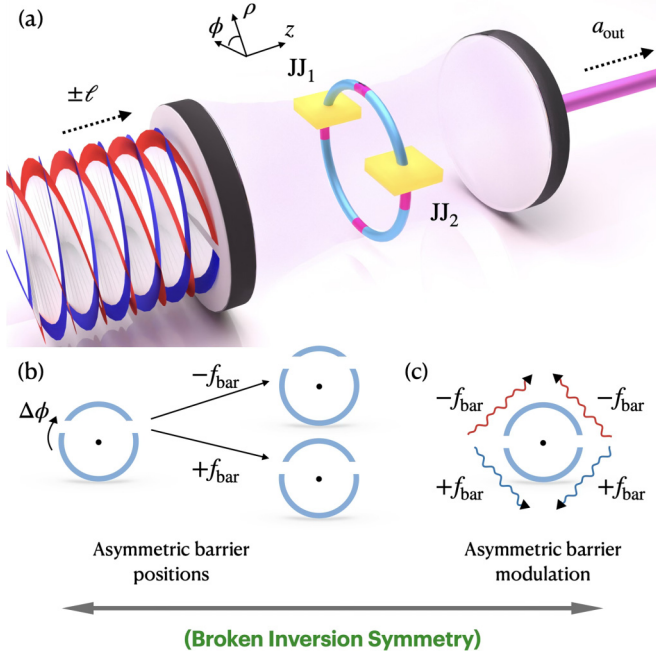


FIG. 1. (a) Schematic diagram of the proposed experimental setup. A ring BEC with two optical barriers [an atomtronic SQUID (AQUID)] is kept at the center of an optical cavity that is driven by the superposition of two LG beams. The output optical field from the cavity is denoted by a_{out} . (b) Schematic protocol to break the inversion symmetry of the condensate by placing the junction such that it divides the ring into two unequal arms. Here, $\Delta\phi$ quantifies deviation from the diametrically opposite positions. (c) Schematic protocol to break the spatiotemporal symmetry by an asymmetric ac drive to the initially symmetric junctions.

junctions move toward the shorter arm. This results in non-reciprocal critical currents ($|I_{c+}| \neq |I_{c-}|$), thereby realizing a position-tunable Josephson diode. In the second configuration, biharmonic modulation of the junction positions, with a nonzero phase difference between the two tones, breaks the spatiotemporal symmetry of the system, enabling a drive-tunable diode [50,51].

Physical setup. Our proposed experimental setup consists of a ring BEC of ^{23}Na atoms, placed at the center of a high-finesse optical cavity, as shown in Fig. 1(a). By denoting the radius of the ring trap as R_0 , and the radial and axial harmonic trapping frequencies as ω_ρ and ω_z , respectively, the trapping potential can be expressed as $U(\rho, z) = \frac{1}{2}m\omega_\rho^2(\rho - R_0)^2 + \frac{1}{2}m\omega_z^2z^2$, where m is the mass of each atom. As a result of tight confinement along the radial and axial directions, atoms remain frozen in the ground state along those directions throughout the evolution, and this makes the azimuthal degree of freedom along ϕ as the only relevant dynamical variable. Therefore, the system can be reduced to an effective one-dimensional description, which is accessible with current experimental capabilities [52] and has been successfully applied to model experiments [43], with the condition on the total number of atoms, N , as [53,54]

$$N < \frac{4\sqrt{\pi}R_0}{3a_s} \left(\frac{\omega_\rho}{\omega_z} \right)^{\frac{1}{2}}, \quad (1)$$

where a_s is the s -wave scattering length of the atoms.

We analyze the Josephson dynamics by moving the JJs towards each other with velocity f_{bar} up to time t_{bar} . This results in a dc supercurrent (ac current) if $f_{\text{bar}} < f_c$ ($f_{\text{bar}} > f_c$), where, f_c is the critical barrier velocity [44,45]. To probe the Josephson oscillation frequency, we drive the cavity with a superposition of two Laguerre-Gaussian (LG) beams with orbital angular momentum (OAM) $\pm\ell\hbar$, which form an angular optical lattice potential about the cavity axis [55]. These beams are blue detuned far from any atomic transition and interact weakly with the condensate atoms. After the light-matter interaction, the cavity output field carries the signature of BEC dynamics (i.e., oscillation frequencies) and can be measured by the homodyne detection technique of cavity optomechanics [56].

Theoretical model. To numerically simulate the above physical system, we describe the azimuthal motion of BEC by a one-dimensional (1D) Hamiltonian [54,57,58],

$$H = \int_0^{2\pi} d\phi \Psi^\dagger(\phi) \left[\mathcal{H}^{(1)} + \frac{g}{2} \Psi^\dagger(\phi) \Psi(\phi) \right] \Psi(\phi) - \hbar \Delta_o a^\dagger a + i \hbar \eta_p (a^\dagger - a), \quad (2)$$

where $\Psi(\phi)$ and a are the bosonic and photonic operators that obey $[\Psi(\phi), \Psi^\dagger(\phi')] = \delta(\phi - \phi')$ and $[a, a^\dagger] = 1$, respectively. $\mathcal{H}^{(1)}$ represents the single-atom Hamiltonian, $\mathcal{H}^{(1)} = \frac{\hbar^2}{2mR_0^2} \left(-i \frac{d}{d\phi} - \Omega \right)^2 + \hbar U_0 \cos^2(\ell\phi) a^\dagger a + V_{\text{bar}}$. Here, the first term denotes the rotational kinetic energy of the atoms, the second the optical lattice potential with strength U_0 (see Supplemental Material [59] for a detailed derivation), and the third the optical barrier potentials, modeled as $V_{\text{bar}}(\phi, \tau) = \sum_{j=1}^2 V_j \exp[-2(\frac{\phi - \phi_{bj}(\tau)}{\sigma_j})^2]$, with barrier height V_j , width σ_j , and barrier position ϕ_{bj} . The second term in Eq. (2) refers to the two-body atomic interaction energy with strength g , and the last two terms denote the energy contributions from the cavity field and laser drive, respectively. $\Delta_o = \omega_L - \omega_0$ denotes the drive-cavity detuning and $\eta_p = \sqrt{P_{\text{in}}\gamma_0/\hbar\omega_0}$ is the pump rate in terms of input optical power P_{in} and cavity decay rate γ_0 .

Using a mean-field description and taking into account thermal fluctuations (stochastic Gross-Pitaevskii equation) and laser shot noise, we describe the temporal evolution of the condensate wave function ψ and the light-field amplitude α as [54,56,57,60,61]

$$(i - \Gamma) \frac{d\psi}{d\tau} = \left[\left(-i \frac{d}{d\phi} - \Omega' \right)^2 + \frac{U_0}{\omega_\beta} |\alpha(\tau)|^2 \cos^2(\ell\phi) + V_{\text{bar}}(\phi, \tau) + \mathcal{G} |\psi(\phi, \tau)|^2 \right] \psi + \xi(\phi, \tau), \quad (3)$$

$$i \frac{d\alpha}{d\tau} = \left\{ - \left[\Delta_c - U_0 \langle \cos^2(\ell\phi) \rangle_\tau + i \frac{\gamma_0}{2} \right] \alpha + i \eta_p \right\} \omega_\beta^{-1} + i \sqrt{\gamma_0} \omega_\beta^{-1} \alpha_{\text{in}}(\tau), \quad (4)$$

using scaled energy $\hbar\omega_\beta = \hbar^2/2mR_0^2$, time $\tau = \omega_\beta t$, rotation rate $\Omega' = \Omega/2\omega_\beta$, atomic interaction strength $\mathcal{G} = g/\hbar\omega_\beta$, and a phenomenological damping term Γ . Previously, a similar description has been successfully used to model BEC-cavity experiments and they have shown excellent agreement

with the mean-field simulation results [62,63]. Interaction with the external environment is taken into account through adding the stochastic terms to Eqs. (3) and (4) that denote the condensate's thermal fluctuations and the laser shot noise, respectively, and are described by δ -correlated white noise, with correlations [54,60]

$$\langle \xi(\phi, \tau) \xi^*(\phi', \tau') \rangle = \frac{2\Gamma k_B T}{\hbar\omega_\beta} \delta(\phi - \phi') \delta(\tau - \tau'), \quad (5)$$

$$\langle \alpha_{\text{in}}(\tau) \alpha_{\text{in}}^*(\tau') \rangle = \omega_\beta \delta(\tau - \tau'), \quad (6)$$

where k_B is the Boltzmann constant and T is the BEC temperature.

Simulation details. We generate the ground state of the condensate in the presence of optical barriers using the imaginary-time evolution method [57]. For simulating the barrier movement step, their positions are modeled as $\phi_{b1,2} = \pm(2\pi f_{\text{bar}})t$ in the rotating frame, up to time t_{bar} . Subsequently, we solve the coupled BEC-cavity equation [Eqs. (3) and (4)] for a range of barrier velocities up to time t_{sim} , which allows us to characterize the superfluid (dc) and dissipative (ac) regime from the power spectrum of phase quadrature of the cavity output field ($S(\omega) = |\text{Im}[\alpha_{\text{out}}(\omega)]|^2$). The cavity output field $\alpha_{\text{out}}(t)$ is related to the intracavity field and optical fluctuations through $\alpha_{\text{out}} = -\alpha_{\text{in}} + \sqrt{\gamma_0}\alpha$ [56]. The cavity field is activated when the barrier motion ceases, ensuring that it does not capture any transient dynamics during barrier motion. This procedure yields the same results, though in real time, *in situ*, and nondestructively, as the methods of detecting Josephson dynamics via absorption imaging of the condensate once the barrier motion stops [44,46,47,64].

Detection of the diode effect. We next focus on achieving the position-tunable diode. Figures 2(a) and 2(b) show the temporal evolution of the atomic tunneling current as the barriers move with velocities $\pm f_{\text{bar}}$. When the barriers are symmetrically placed across the diameter of the ring ($\Delta\phi = 0$), equal and opposite barrier velocities generate equal and opposite dc supercurrents around the ring. However, for asymmetrical placed barriers [see Fig. 1(b)], a positive bias (toward the longer arm) produces a dc current, while the same negative bias (toward the shorter arm) leads to oscillation in the current. This behavior indicates that the system has entered into the dissipative regime for negative biasing, while it remains superfluid for positive biasing. In our setup, this discrepancy is detected via the power spectrum of the phase quadrature of the cavity output field $S(\omega)$, which characterizes the dc (ac) Josephson regime by zero (nonzero) splitting of all the side-mode peaks (see Supplemental Material [59] for a detailed discussion on dc-ac transition), and the magnitude of splitting $\Delta\omega$ is the same as the Josephson oscillation frequency [49].

We have presented the cavity output spectrum $S(\omega)$ for the symmetric and asymmetric cases in Figs. 2(c) and 2(d), respectively. In the symmetric case, the spectrum remains the same and exhibits no splitting of side modes for both positive- and negative-current biases. However, in the asymmetric case, the spectrum shows side-mode splitting for negative biasing but not for positive biasing. This supports the findings from Fig. 2(b) that the dc current flows for only positive biasing for the chosen barrier velocity. We plot the magnitude

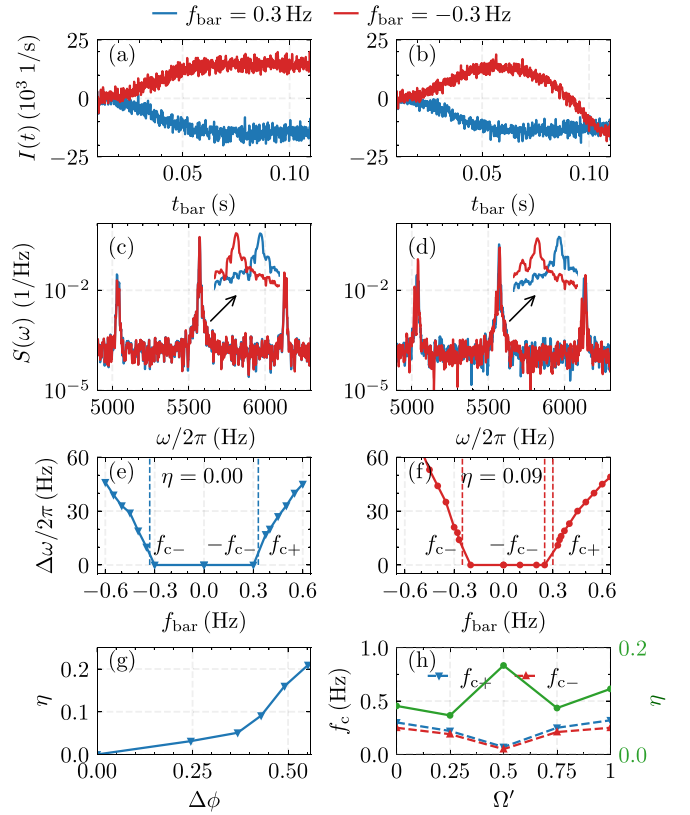


FIG. 2. Position tunable diode: (a), (b) Josephson tunneling current during the junction movement period with frequency $f_{\text{bar}} = \pm 0.3$ Hz for (a) initially symmetric barriers ($\Delta\phi = 0$) and (b) initially off-centered barriers ($\Delta\phi = 0.43$). (c), (d) Power spectrum of phase quadrature of cavity output field corresponding to the case of (a) and (b), respectively. In the inset, the curves are deliberately displaced horizontally for clarity. (e), (f) Magnitude of peak splitting in the cavity output spectrum vs barrier velocity for (e) $\Delta\phi = 0$ and (f) $\Delta\phi = 0.43$. The vertical dashed lines indicate the critical barrier velocity, obtained via fitting: $\Delta\omega = A\sqrt{f_{\text{bar}}^2 - f_c^2}$. (g) Variation of diode efficiency η with the amount of initial asymmetry $\Delta\phi$. (h) Variation of critical barrier velocity f_c and η vs AQUID rotation rate Ω' . The other parameters used are $N = 3700$, $R_0 = 4 \mu\text{m}$, $V_0 = 17.5 \mu_0 = 25$ nK, $\sigma = 0.7 \zeta = 1.8 \mu\text{m}$, $\zeta = 1/\sqrt{\mu_0/\hbar\omega_\beta}$, $t_{\text{bar}} = 0.1$ s, $T = 10$ nK, $P_{\text{in}} = 10$ fW, $\Gamma = 0.0001$, $U_0 = 2\pi \times 212$ Hz, $\bar{\Delta} = \Delta_0 - U_0 N/2 = -2\pi \times 173$ Hz, $\Delta_a = 2\pi \times 4.7$ GHz, $\omega_0 = 2\pi \times 10^{15}$ Hz, $\omega_\rho = \omega_c = 2\pi \times 42$ Hz, $\gamma_0 = 2\pi \times 2$ MHz, and $\Omega' = 0$.

of splitting $\Delta\omega$ versus barrier velocity f_{bar} , a current (I)-voltage (V) curve analog, for both cases in Figs. 2(e) and 2(f). The critical velocity is extracted via fitting the data with $\Delta\omega = A\sqrt{f_{\text{bar}}^2 - f_c^2}$, where the prefactor A is proportional to the circuit resistance R [47]. Details on detecting the dc-ac transition and critical velocity are provided in Supplemental Material [59]. The critical velocities in the positive and negative directions are denoted by f_{c+} and f_{c-} , respectively. For the symmetric case, the I-V curve is also symmetric and the critical velocities in both directions are the same. In the other case, f_{c+} and f_{c-} differ by a nonzero amount, which signifies that we achieve a diodelike behavior for particular barrier velocities ($|f_{c-}| < |f_{\text{bar}}| < |f_{c+}|$) when the system remains in the superfluid regime (dc current) for positive biasing and in

the resistive regime (ac current) for equal negative biasing. We quantify this difference in critical barrier velocities as the diode efficiency factor, $\eta = \frac{|I_{c+}| - |I_{c-}|}{|I_{c+}| + |I_{c-}|} = \frac{|f_{c+}| - |f_{c-}|}{|f_{c+}| + |f_{c-}|}$ [21,23,25].

The physical origin of this diode effect is related to the way the phase difference across the barriers, $\Phi = \Phi_1 - \Phi_2$, evolves when the barriers are in motion. Moving barriers develop a density imbalance leading to a nonzero chemical potential bias. For low barrier velocities, atoms tunnel through the barriers and cancel the bias. As a result, the phase difference Φ remains constant, giving rise to a dc supercurrent. At higher velocities, the density imbalance builds up so quickly ($\Delta\mu \neq 0$) that the phase difference cannot stay constant and starts running, $\frac{d\Phi}{dt} = -\frac{\Delta\mu}{\hbar}$, and the current starts oscillating as $I = I_c \sin(\Phi)$. When the barriers start from an asymmetric position, the density imbalance builds up more rapidly in the shorter arm than in the longer arm. So the dc-to-ac transition occurs earlier when barriers are moved towards the short arm (negative biasing) instead of towards the long arm (positive biasing), yielding unequal critical currents $|I_{c+}| \neq |I_{c-}|$. Therefore, by increasing the initial asymmetry, we achieve higher diode efficiency [see Fig. 2(g)]; hence, a position-tunable diode is realized. Due to quantum interference of clockwise and counterclockwise currents generated from barrier motion, critical currents oscillate periodically with a minimum at half-integer value of scaled rotation rate Ω' [49,64]. This oscillation further allows the tunability of the diode efficiency, as shown in Fig. 2(h).

Applications. First, we demonstrate the emergence of asymmetric Shapiro steps in the current-chemical potential difference curve ($f_{\text{bar}} - \Delta\omega$ curve) by applying an ac current in the form of periodically driven barriers. In simulation, this motion is modeled as $\phi_{b1,2} = \pm(2\pi f_{\text{bar}})t + \phi_m \sin(2\pi f_m t)$. Figure 3(a) shows that the plateau in the chemical-potential difference, which is quantized by the external barrier modulation frequency f_m , is symmetric under positive and negative barrier velocities for a symmetric configuration of the barriers. In the other case, when the barriers are placed asymmetrically, the Shapiro step heights are quantized, but the location and width of the plateau are different for positive and negative biasing, as seen in Fig. 3(b). Therefore, for a sufficiently large asymmetry, the Shapiro step can vanish for a particular biasing direction, leading to a nonreciprocal barrier modulation detector.

We demonstrate the use of the atomtronic Josephson diode as a half-wave rectifier in Figs. 3(c) and 3(d). We achieve this functionality by applying a square-wave bias current in the form of barrier motion with alternating direction up to three cycles. First, we move the barriers towards the longer arm of the ring with velocity $+f_{\text{bar}}$ to induce a positive current, and then reverse the barrier motion with the same magnitude ($-f_{\text{bar}}$) to induce the negative bias current. The corresponding current profile is shown in Fig. 3(c). During the barrier motion period, we keep the cavity field off and turn it on once the barrier motion stops to measure the chemical-potential difference. Between successive intervals of turning off and turning on the cavity field, the barriers move for $t_{\text{bar}} = 0.1$ s, which is long enough for the cavity field to decay (within $\sim \mu\text{s}$). Hence, each measurement is independent, not carrying information from the previous measurement. To obtain the rectification, we have used $|f_{c-}| < |f_{\text{bar}}| < |f_{c+}|$, and the corresponding

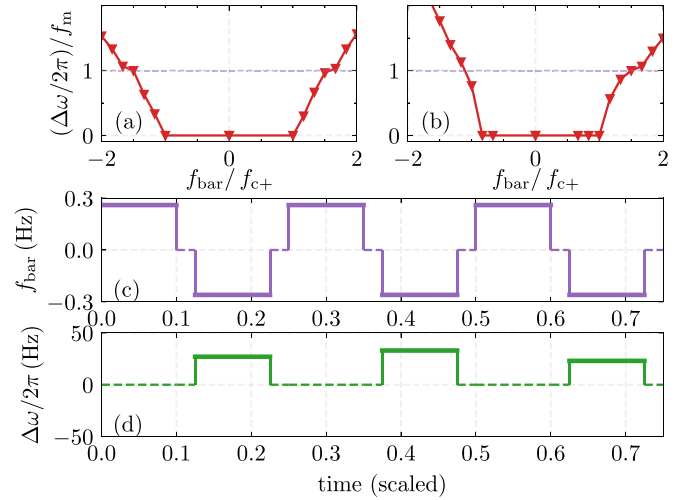


FIG. 3. Applications of atomtronic Josephson diode: (a), (b) Direction-dependent modulation detector. Variation of the magnitude of peak splitting ($\Delta\omega$) vs barrier velocity (f_{bar}) for $f_m = 30$ Hz, $\phi_m = 0.01$, where (a) $\Omega' = 0$, $\Delta\phi = 0$ and (b) $\Omega' = 0$, $\Delta\phi = 0.43$. (c), (d) Half-wave rectifier. (c) Three cycles of square wave of bias current that correspond to $f_{\text{bar}} = \pm 0.26$ Hz (lies in between f_{c+} and f_{c-}). (d) Measured junction voltage ($\Delta\omega$). In the horizontal axes of (c) and (d), t_{sim} is scaled down to $t_{\text{bar}}/4$ for a clear demonstration of current cycles. Here, $t_{\text{bar}} = 0.1$ s, $f_m = 0.0$ Hz, $\phi_m = 0.0$, $\Omega' = 0$, and the other set of parameters are the same as used in Fig. 2.

chemical-potential difference ($\Delta\omega$) for each measurement cycle is shown in Fig. 3(d). This yields a zero chemical-potential difference (“off” state) for the positive current but a nonzero value (“on” state) for the negative current. Hence, the bias current in the form of a square wave is rectified. However, the cycle count is limited by the lifetime of the condensate and the duration of the barrier motion. Additionally, a careful selection of barrier velocity that lies between the critical values is necessary to achieve rectification.

Drive-tunable diode. We discuss the realization of a Josephson diode by using a biharmonic ac drive. This breaks the spatiotemporal symmetry of the system when the phase difference between the two harmonics is $\theta \neq n\pi$, where n is an integer. This phase difference makes the ac drive asymmetric (unequal maximum and minimum), which leads to different barrier motions in forward and backward directions [see Figs. 4(a) and 4(c)], resulting in unequal critical currents in the two directions. In simulation, the asymmetric biharmonic ac drive is achieved via modeling the barrier motion as [50,51]

$$\phi_{b1,2} = \pm(2\pi f_{\text{bar}})t + \phi_m [\sin(2\pi f_m t) + \sin(4\pi f_m t + \theta)]. \quad (7)$$

The resulting barrier velocity versus chemical-potential difference curve (I-V curve) is shown in Fig. 4(d), which clearly reveals a difference in the critical velocity magnitude for the forward $+f_{\text{bar}}$ and backward $-f_{\text{bar}}$ directions, leading to a nonzero diode efficiency of 91%. Additionally, the I-V curve exhibits asymmetric Shapiro steps, a characteristic feature of the JDE [21,28,50]. As predicted by Eq. (7), reversing the sign of both f_{bar} and θ results in identical barrier motion

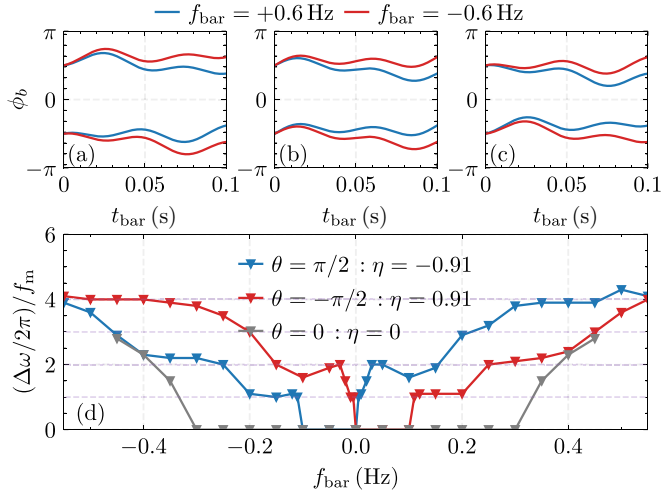


FIG. 4. Drive-tunable diode: temporal evolution of barrier positions for $f_m = 10$ Hz, $\phi_m = 0.22$, with (a) $\theta = \pi/2$, (b) $\theta = 0$, and (c) $\theta = -\pi/2$. (d) Variation of the magnitude of peak splitting ($\Delta\omega$) vs barrier velocity (f_{bar}). The other set of parameters used here is the same as used in Fig. 2(a).

and, consequently, the same peak splitting magnitude $\Delta\omega$, in the cavity output spectrum. This symmetry allows for easy flipping of the diode efficiency's sign by simply changing the sign of θ , while maintaining the same efficiency magnitude. Such a feature enables real-time switching and current rectification with minimal operational requirements. We have demonstrated half-wave rectification of a square-wave bias current signal using this setup, along with the corresponding cavity output spectrum, in Supplemental Material [59].

Conclusion and future scope. We have proposed and simulated the realization of the Josephson diode effect in an atomtronic circuit by using asymmetrically placed optical barriers or an asymmetric ac drive. They break the spatiotemporal symmetry of the system, giving rise to unequal critical currents in the forward and backward directions. The diode efficiency is quantified by measuring the difference between the magnitudes of critical currents. We have demonstrated a maximum diode efficiency of 91%, which can be further tuned by the barrier and drive parameters. The capability of this atomtronic Josephson diode is demonstrated through the rectification of a square-wave signal, whose rectification cycle can be further increased by reducing the barrier movement and measurement time. Future work will explore the Josephson diode effect in other circuit geometries, including non-Hermitian junctions, and with an efficiency of 100%. We expect that our one-dimensional predictions will be robust to higher dimensionality, following similar works [43,65,66]. Additionally, the cavity-based measurement technique demonstrates the potential of cavity optomechanics as a single-shot, highly sensitive probe for studying condensate dynamics in atomtronic Josephson circuits.

Acknowledgments. We gratefully acknowledge the supercomputing facilities Param Ishan and Param Kamrupa, where all the simulation runs were performed. M.B. thanks the Air Force Office of Scientific Research (Grant No. FA9550-23-1-0259) for support. R.K. was supported by JSPS KAKENHI (Grant No. 25K07190) and JST ERATO (Grant No. JPM-JER2302).

Data availability. The data that support the findings of this article are not publicly available. The data are available from the authors upon reasonable request.

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